# Multimodal Remote Sensing Image Matching Combining Learning Features and Delaunay Triangulation

Yongxian Zhang, Yuxuan Liu, Haiming Zhang, and Guorui Ma

***Abstract*— Multimodal remote sensing image (MRSI) high-precision matching faces significant challenges due to the large nonlinear distortions in radiation and geometry. To address this problem, we propose a novel MRSI matching method, multimodal remote sensing image matching with Delaunay triangulation constraint network (M2DT-Net), which takes advantage of deep convolutional neural networks (CNNs) and the Delaunay triangulation (DT) strategy. First, a fine-tuned CNN model is applied to learn invariant features with high robustness across MRSIs. Second, an adaptive feature descriptor distance constraint combined with the DEGENSAC refinement is employed to filter the obtained matches, and only the matches with high credibility are retained. Finally, a DT strategy is adopted to expand more underlying correct matches that are dropped inliers in the previous stage. The experimental results on seven types of MRSI pairs indicate that M2DT-Net is superior to seven state-of-the-art image matching algorithms in terms of model performance and efficiency, achieving a balance of robustness and time consumption. The image registration results further demonstrate the effectiveness of the proposed algorithm. Therefore, the proposed M2DT-Net method provides a reference for resolving common multimodal image matching problems.**

***Index Terms*— Deep learning, Delaunay triangulation (DT), model fine-tuning, multimodal remote sensing image (MRSI) matching.**

## I. Introduction

With the rapid development of high-resolution Earth observation technology, remote sensing images acquired by multisensor and multimodal technology are becoming more abundant. Imaging cameras range from panchromatic cameras to multispectral and hyperspectral cameras, and imaging platforms range from visible to nightlight, infrared, microwave, and light detection and ranging (LiDAR), realizing multisource collaborative observation. The technology has been developing toward high spatial, high temporal, and high spectral resolution data acquisition, which provides an important data source for remote sensing image information deep mining and big data analysis, as shown in Fig. 1. Compared with single-source sensor observation data, the information provided by multisource remote sensing images is complementary, which compensates for the limitation of single sensor observations. More reliable results are obtained in many applications, such as image retrieval [1], image registration [2], and disaster assessment [3]. These application scenarios need to closely align images on the same coordinate system [4]. The key step is feature matching and obtaining matching points between two multimodal remote sensing images (MRSIs). However, there are significant differences in the appearance of multimodal images due to the different imaging geometrics and physical radiation mechanisms. Various sensors, viewpoints, scales, illumination times, and textures may cause feature keypoint confusion and object occlusion problems. On the other hand, variations in a camera's internal parameters could differ in an image quality and imaging model. These factors make it more difficult to match multisource images with high precision. Therefore, it is crucial to explore a more robust, effective, and reliable MRSI matching algorithm.

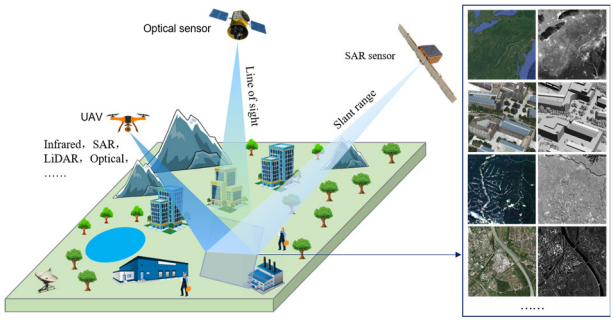



Fig. 1. Comparison of MRSIs.

To address the MRSI matching problem, many excellent methods have been developed; they can be roughly classified into handcrafted methods and deep learning methods [5]. The main difference is that these methods rely on different

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prior knowledge and the availability of informative feedback. Feature detectors are divided into three types: corner, edge, and region. Whether handcrafted features or deep learning features, a good feature detector should have three important properties [6]: 1) it can effectively discover the key discriminative invariant information in task images; 2) it can repeatedly detect consistent features when the test scene undergoes diverse transformations; and 3) it should be as efficient as possible.

Handcrafted methods focus on the use of image low-level information, such as corner, gray, texture, and edge features, to fulfill the MRSI matching task. HOPC [7] adopts the phase congruency model to yield consistent orientation information and to capture image structural similarity. Radiation-variation insensitive feature transform (RIFT) [8] builds a radiation-variation insensitive feature matching method that performs well on MRSIs. Xiong et al. [9], [10], [11] used the image local self-similarity feature to achieve MRSI high-precision matching and registration. CoFSM [12] constructs a new image scale space for increasing the number of corresponding points and involves a low-pass filter to enhance feature robustness across MRSIs. These handcrafted methods achieved good matching results in their experimental MRSI data. However, we discovered that the performance becomes unstable with significant differences in rotation, viewpoint, and scale, so they do not work well with complex real-world backgrounds.

Deep learning methods, as data-driven models, offer greater flexibility than traditional handcrafted matching approaches. As image high-level feature detectors, these methods perform local adaptive learning image information from low level to high level and learn robust features and descriptors from the training dataset by multilevel nonlinear mapping. Therefore, deep learning methods can be well applied to multimodal image matching [13]. In research on learning feature matching, many excellent matching algorithms have been proposed in the fields of computer vision and remote sensing. In computer vision, DELF [14] achieves robust matching and geometric consistency verification via weak supervision training and incorporates a self-attention mechanism for high-level features. Key.Net [15] extends the covariance constraint to a multiscale feature detector to achieve better matching. SuperPoint [16] designs a fully convolutional neural network (CNN) architecture by operating on a full-sized image and jointly computing pixel-level interest point locations and associated descriptors in a forward pass. D2-Net [17] presents a describe-and-detect approach to obtain local learning feature detection and description, with a dual role in both detection and description, showing great power for solving image matching problems in diverse scenes. R2D2 [18] introduces an unsupervised means to learn feature detectors and local descriptors, which are both repeatable and reliable. These deep learning methods have achieved state-of-the-art matching results in familiar digital camera images. Nevertheless, it is difficult to establish a subpixel corresponding relation between MRSIs due to the large differences in nonlinear radiometric and geometric distortions. Several researchers in the field of remote sensing have explored how to effectively match MRSIs using deep learning methods. Lan et al. [19] proposed CMM-Net by fine-tuning the D2Net model, achieving good adaptability in MRSI matching. MAP-Net [20] achieves multimodal synthetic aperture radar (SAR) and optical image efficient and robust matching via an image-based CNN with an attention mechanism and spatial pyramid aggregated pooling strategy. Tharani et al. [21] designed a descriptor network by unsupervised encode–decode features, which obtains robust performance in MRSI matching and image retrieval applications. For MRSI matching, due to the lack of sufficient training data in the target task, it is quite difficult to significantly improve the model performance by training a learnable model only on a small training dataset. Fortunately, fine-tuning is the most widely used approach to improve the model performance; it starts with a pretrained model on the source task and trains it further on the target task. Zhang et al. [4] achieved high-precision matching of optical and SAR images by fine-tuning the pretrained model.

In summary, multimodal image matching methods based on traditional handcrafted features and deep learning features have made great progress. However, as the depth and breadth of MRSI application fields continue to expand, higher demands are placed on them, and there is still some room for improvement. The main manifestations are described as follows.

1) The existing MRSI matching algorithms have poor performance for images with large affine transformations.

2) A sufficient training dataset is an important approach to improving the model performance. However, there are few public MRSI training datasets, which limits the application of image matching.

3) There is a lack of robust MRSI feature detector methods with an abundant number of features, uniform distribution, and high repetition rates.

To address the above three key problems, a novel end-to-end matching method is proposed to effectively match the MRSI with the large difference in nonlinear geometric and radiation distortion. The algorithm uses a fine-tuned pretrained learning feature detector model to extract and describe the key features that are suitable for MRSI matching tasks and constructs a Delaunay triangulation (DT) constraint to expand the matching inliers and their distribution. The experimental results of eight different scenario datasets show that our proposed algorithm achieves high-precision MRSI matching. The proposed method is referred to as multimodal remote sensing image matching with Delaunay triangulation constraint network (M2DT-Net). Compared with existing algorithms, the main contributions of this article are listed as follows.

1) To improve the robustness of the feature detector model, a fine-tuned end-to-end network model is utilized to mitigate the domain gap between two MRSIs, extract invariant features with abundant quantities and uniform distributions, and enhance the image matching performance.
2) To retain more discriminative information in image features, a dynamic adaptive distance constraint method is designed to filter out outliers and retrain high-quality matched points.
3) To obtain more abundant correct corresponding points, a DT constraint method is constructed to expand

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more underlying corresponding points under complex real-world background conditions. The experimental results show that the proposed M2DT-Net method outperforms state-of-the-art multimodal image matching methods in terms of the number of correct matches (NCMs), accuracy, and efficiency.

The remainder of this article is organized as follows. Section II reviews the related work on network model fine-tuning technology and DT constraint methods. Section III introduces the proposed MRSI matching method in detail. Section IV shows the test datasets, results, and qualitative and quantitative analysis. The conclusion is provided in Section V.

## II. Related Work

### A. CNN Model Fine-Tuning

Fine-tuning a pretrained network model on a new dataset is the most common strategy for knowledge transfer in the context of deep learning [22]. Fine-tuning the pretrained model on the target dataset could reduce the gap between the source dataset and the target dataset [23]. Therefore, fine-tuning is an effective approach to learning from limited label datasets [24]. Many methods have been proposed to fine-tune network models, such as fine-tuning the parameters of the last few layers or just using the pretrained model as a fixed feature detector. Compared with training from scratch, fine-tuning a pretrained model on a target dataset can dramatically improve the model performance while reducing the target labeled data requirements [25]. Fine-tuning has an important role in many image processing fields.

In computer vision, fine-tuning pretrained models have been proven to be powerful in a variety of problems. It is a common practice to work with pretrained models [24]. Salman et al. [26] showed that fine-tuning from an adversarial pretrained model can outperform a standard pretrained model. Lecoutre et al. [27] managed to achieve a higher performance for the classification task by fine-tuning the deeper ResNet50 model. Shin et al. [28] adopted a fine-tuned pretrained model to map images. It can be seen that the fine-tuning model has a better performance than the existing models and training from scratch in learning feature extraction [29].

In medical image analysis, fine-tuning a pretrained model is an effective scheme to reach the best model performance for the application based on the amount of available data. Margeta et al. [30] fine-tuned a pretrained model to retrieve key information from magnetic resonance imaging. Tajbakhsh et al. [31] showed that fine-tuned CNNs were more robust to the size of training sets than CNNs that were trained from scratch. After fine-tuning the network model, image low-level features are maintained from one dataset to another dataset, while image higher level feature parameters are updated to adapt the network to the new task [32]. Therefore, fine-tuning models can be designed to focus on reducing biases in novel class feature distributions [33].

In MRSI matching, it is difficult to collect and label so much training data, making the "big data" required for model training a "few-shot learning" problem. To address such problems, the consensus of the AI community is to use a model pretrained on large public datasets as the backbone of downstream tasks and then fine-tune the model rather than train it from scratch [34]. To the best of our knowledge, a few studies improve image matching performance with MRSI datasets by fine-tuning a pretrained network model. In this study, we fine-tuned the last few layers of the D2-Net network while freezing the parameters of the remaining initial layers to the pretrained values. We extract learning features based on a fine-tuning pretrained model to enhance the MRSI feature representational performance.

### B. DT Constraint Match Expansion

With a rapid increase in remote sensing image resolution, the similarity of neighbor features on the image increases. To be matched, keypoints on the sensed image may have multiple similar feature keypoints on the reference image, which greatly increases the difficulty of high-precision image matching [35].

To the best of our knowledge, classical matching methods, such as force matching and k-nearest neighbor matching, calculate the similarity between all feature descriptors of two images and then set a threshold to constrain the similarity score to fulfill feature matching [36]. However, the distribution of and the number of matched feature points are difficult to determine. Previous studies have shown that a DT has a simple structure and strong stability characteristics. A DT can express the image information as topological relationships through the matched keypoints, split the image plane into reliable transformations between two corresponding triangles, and restrict the searching space for candidate matches [37]. The DT can be used to obtain initial matches based on the properties of dynamic generation and good geometry. The DT vertices consist of keypoints, and edges are added according to the empty circle property. As a result, the DT can improve the matching model robustness for nonlinear distortion between two multimodal images. Zhu et al. [38] adopted the DT constraint to achieve match propagation from reliable matched corresponding points, which had a role identical to the match expansion. Subsequently, some similar works have been documented using DT constraints to expand matching points. Zhu et al. [39] employed a triangle constrained candidate searching strategy to obtain more potential matching points in multiple close-range image matching. Wu et al. [40] integrated seed points and seed edges and constructed a pair of initial edge-constrained triangulations to expand candidate matches and significantly improve match reliability on poor textural images. Paul and Pati [41] adopted the improved SAR-SIFT algorithm and DT local matching to increase the controllability of the number of extracted features and even the distribution of the feature points, significantly improving the model performance. In multispectral/multidate remote sensing image registration, Zhao et al. [42] proposed a robust DT matching algorithm to achieve reliable registration of various representative images. Zhang et al. [43] proposed an efficient remote sensing image registration algorithm based on the DT constraint. These methods show that additional candidate keypoints can be obtained to expand matching

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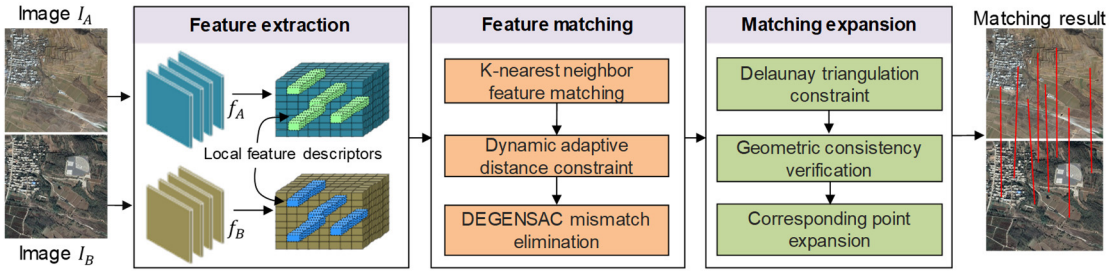



Fig. 2. Workflow of the proposed method.

points using the DT constraint strategy. Although the feature matching keypoint expansion method by the DT constraint has been reported in some studies, our M2DT-Net has many different considerations and scenarios.

### C. Relation to Previous Works

The proposed method is closely related to MRSI matching in various real-world scenarios. The difference from the existing image matching method is that we try to incorporate the fine-tuned pretrained model, adaptive distance matching, and the DT constraint strategy into feature matching to increase the number of corresponding points and the uniformity of the matches distribution to achieve reliable and robust MRSI high-precision matching. In contrast, previous methods mainly focus on specific types of matching, while the proposed method meets the needs of image matching in complex backgrounds.

## III. Methodology

### A. Workflow of the Proposed Method

This section introduces the proposed M2DT-Net algorithm in detail, including the three steps of feature extraction, feature matching, and matching expansion.

The diagram of the M2DT-Net architecture, shown in Fig. 2, depicts the following processes.

1) The first module is feature extraction, which is performed to obtain local learning feature maps of input image pairs $I_A$ and $I_B$. In this step, learning features are extracted based on fine-tuning the pretrained model.
2) The second module is feature matching, which is aimed at obtaining reliable matching keypoints. This step will adopt the k-nearest neighbor Euclidean distance matching method to establish the corresponding keypoints between two local feature descriptors. This module retains the $k$ matching keypoints with the highest confidence score to remove outliers and to reduce the matching time to obtain sparse local features. Afterward, the dynamic adaptive distance constraint and DEGENSAC [44] mismatch elimination method are used to refine the corresponding points.
3) The third module is matching expansion, which is used to explore additional underlying inliers. This step is aimed at expanding the underlying corresponding points using the constraint between the refined Delaunay triangles and their corresponding graph. The input to the network is two images, and the output to the network is the final matching result.

In Section III-B, we will describe each of the three modules in detail.

### B. Robust Learning Feature Extraction

*1) Feature Extraction Network Architecture With Scale and Rotation Invariance:* Given the image pair $I_A$ and $I_B$, we use a pretrained CNN model as a feature detector to extract interest points. The CNN architecture without fully connected layers takes an input image and produces a feature map $f \in \mathbb{R}^{h\times w\times d}$, which can be interpreted as an $h \times w$ dense spatial grid of $d$-dimensional local descriptors [45]. The proposed CNN architecture is based on the D2-Net model, where we improved the backbone network to extract features with scale and rotation invariances.

To retain high- and low-level local feature information with as much detail as possible, the backbone network uses a fine-tuned VGG16 network, which is cropped at the pool4 layer (before the ReLU unit), followed by the feature $L2$-norm. The improved network is cropped into a $512 \times 512 \times 3$ image region. We fine-tuned the last three layers, kept the earlier layer weights frozen, and output the last conv4_3 feature map of the fourth block. The feature map is employed as the output result of the feature extraction module (Fig. 3).

To obtain multiscale features, we adopt the image pyramid model to construct feature representations. For input image $I$, the image pyramid is composed of $L$ layers, where the first layer represents the original image $\sigma = 1.0$, the subsequent layer is the zoom layer of the first level, and the zoom factor is $f = (1/L)^{(1/L-1)}$. The last level of the pyramid is $1/L$ times the initial image. In this study, the image pyramid level is set to three levels, that is, 0.5, 1.0, and 2.0 times resolution. To accommodate the resolution differences between two images, these levels are applied to extract the feature map $F^l$ at each level resolution, and (1) is applied to concatenate the results of different image pyramid levels. The feature descriptors are extracted from the fusion feature map $\tilde{F}^l$

$$\tilde{F}^l = F^l + \sum_{\gamma<l} F^\gamma. \tag{1}$$

To cumulatively fuse different pyramid-level features, the low-resolution features are interpolated to the same size as the high-resolution feature maps. Feature locations are marked

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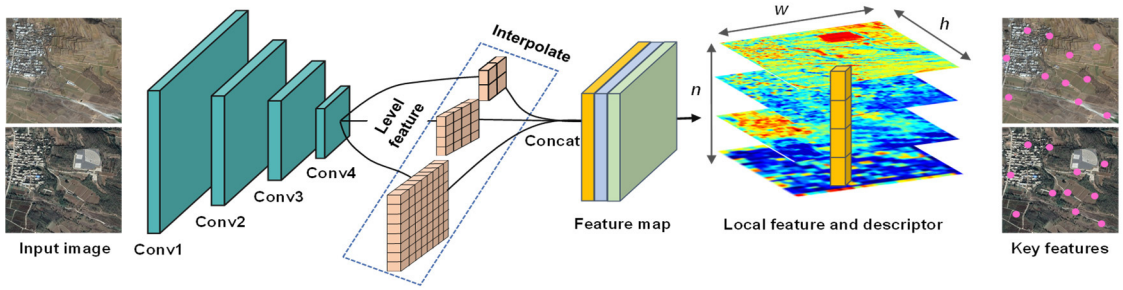



Fig. 3. Architecture of the feature extraction network.

starting from the coarsest scale [17]. The pixel coordinates of feature locations are upsampled to a higher level feature map as a template. If keypoints extracted from a higher level resolution feature map fall into the template, they are disregarded.

To obtain rotation-invariance features, the in-plane image is rotated by angle $\theta$ using a random rotation homography matrix $H_R(\theta)$ when fine-tuning the pretrained model.

*2) Network Training Loss:* The loss function is selected to measure the distance between the model output and the ground truth, indicating the direction for model optimization. Therefore, an appropriate loss function is an important task to improve the model's performance. To fully consider the correspondence between positive matching points and negative matching points, we inherit the training loss function as the detection and description loss in D2Net [17] for its good performance, which is written as

$$L(I_A, I_B) = \frac{1}{|C|} \sum_{c \in C} \frac{s_c^{(A)} s_c^{(B)}}{\sum_{q \in C} s_q^{(A)} s_q^{(B)}} \mathcal{M}\left(d_c^{(A)}, d_c^{(B)}\right) \tag{2}$$

where $C$ is the set of all corresponding points of $(I_A, I_B)$. $s_c^{(A)}$ and $s_c^{(B)}$ are the feature points detection scores, $d_c^{(A)}$ and $d_c^{(B)}$ are their corresponding descriptors, and $\mathcal{M}(\cdot,\cdot)$ is the ranking loss for representation learning, which is written as

$$\begin{aligned} &\mathcal{M}\left(d_c^{(A)}, d_c^{(B)}\right) \\ &= \left[\mathcal{D}\left(d_c^{(A)}, d_c^{(B)}\right) - m_p\right]_+ \\ &\quad + \left[m_n - \min\left(\min_{k \neq c} \mathcal{D}\left(d_c^{(A)}, d_k^{(B)}\right), \min_{k \neq c} \mathcal{D}\left(d_k^{(A)}, d_c^{(B)}\right)\right)\right]_+ \end{aligned} \tag{3}$$

where $[x]_+ = \max(0, x)$ is the hinge loss, and margin $m_p$ and margin $m_n$ are set to 0.2 and 0.1 for positives and negatives, respectively.

*3) Network Model Fine-Tuning:* Training a deep CNN model from scratch is difficult because it requires sufficient labeled training data and considerable expertise to ensure proper convergence [31]. Previous studies have shown that fine-tuning a pretrained model could offer good generalization performance [46]. To address the problem of insufficient training datasets for MRSIs, we fine-tuned the standard VGG16 pretrained model, which is truncated after the Conv4_3 layer to initialize the feature extraction network. In this study, the *Megadepth* and *rotated HPatches* datasets are selected as fine-tuning data to meet multimodal image training. Afterward, features with good similarity are obtained from multimodal images.

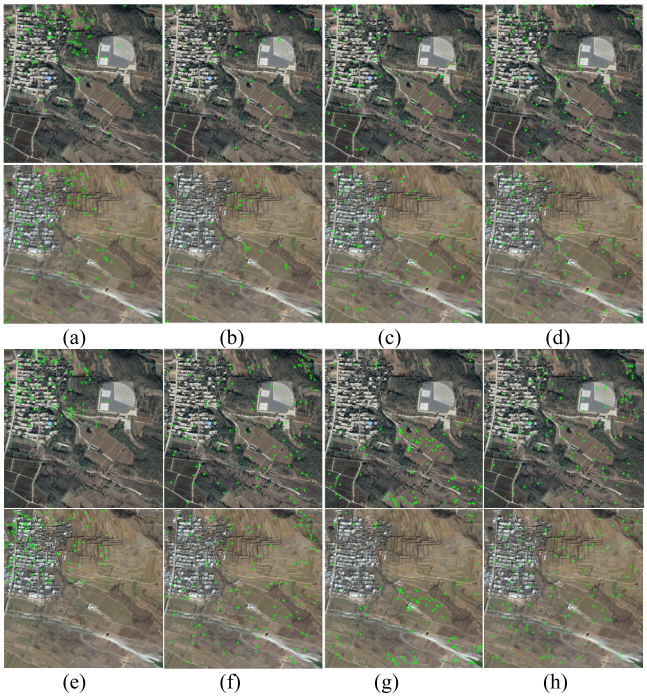



Fig. 4. Comparative results of the top 100 keypoints. (a) RIFT. (b) DELF. (c) D2Net. (d) R2D2. (e) SuperPoint. (f) CMM-Net. (g) RoRD. (h) M2DT-Net.

Fig. 4 shows an example of the top 100 keypoints with the highest scores. Several conclusions are reached from the results: 1) compared with the handcrafted RIFT method, M2DT-Net is robust to different texture information and can obtain more uniformly distributed feature keypoint locations; 2) compared with the DELF, SuperPoint, and RORD learning methods, M2DT-Net can obtain a better matching point distribution; and 3) compared with D2Net, R2D2, and CMM-Net, M2DT-Net can produce dense feature maps in the imagery, indicating that the proposed method is robust to identifying the features.

### C. Feature Matching Under Multiple Constraints

*1) Dynamic Adaptive Distance Constraint for Initial Matches:* High-resolution remote sensing images have local spatial continuity, so there may be several points with

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high similarity near the correct matching keypoints. As a result, the matching confidence is reduced. We adopt the K-nearest neighbor feature matching algorithm to search the corresponding pairs. Because the distance $\text{dis}_j$ of the first corresponding point is smaller than the distance $\text{dis}_j'$ of the second corresponding point, we only retain the top $k$ feature points with the highest correlation coefficient to improve the proposed algorithm performance. Here, $N$ corresponding points are searched, and the average distance difference between the best corresponding point and the second corresponding point is calculated, as in the following equation [19]:

$$\text{avg_dis} < \frac{1}{N}\sum_{j=1}^{N}\left(\text{dis}_j' - \text{dis}_j\right). \tag{4}$$

The outlier elimination condition is that the first distance is less than the difference between the second distance and the average distance difference avg_dis, i.e.,

$$\text{dis}_j < \text{dis}_j' - \text{avg_dis}. \tag{5}$$

The algorithm adopts the average of the distance difference as the criterion to measure feature similarity. The algorithm can retain high-quality corresponding points in the first filtering to improve the stability of feature matching.

*2) Outlier Elimination Based on DEGENASAC:* After the dynamic adaptive minimum distance constraint refinement, many mismatched points are eliminated, and the matching inlier rate is improved. However, there may still be incorrect matches due to large affine transformation, occlusion, and perspective differences between two MRSIs. The latest research demonstrates that the DEGENSAC outlier elimination performance is better than the RANSAC algorithm [47]. The DEGENSAC algorithm can estimate the epipolar geometry and homography more correctly and reliably than RANSAC in the presence of a dominant plane [44]. Following this idea, we use the DEGENSAC for further refinement of the corresponding points. Specifically, a homography matrix is utilized as the transformation model; the confidence is set to 0.9999, the outlier threshold is 3 pixels, the maximum number of iterations is 100 000, and the error type is *sampson*. In addition, the symmetric error check is also utilized, and the keypoints that satisfy the geometric consistency constraints are retained as the correct corresponding points.

### D. Matching Expansion Using DT

Suppose that there is a pair of images $I_A$ and $I_B$ that are to be matched. After the feature extraction and feature matching modules, $n$ matched keypoints $\text{C} = \{(p_i, q_i): i = 1, 2, \ldots, n\}$ are obtained. We can construct the DT $G_A$ and its corresponding graph $G_B$ based on the matches (Fig. 5). However, some correct matching feature keypoints are filtered out, which affects the reliability of the corresponding results, as shown in the red rectangle.

To obtain more reliable corresponding points, we apply the DT technology to identify underlying matching points to expand the number of matches, because it is stable, unique to a given point set, and maximizes the minimum angle. Nonetheless, for matching expansion based on the DT constraint, there are two problems to consider: 1) how can prevent mismatching points between two Delaunay triangle meshes? and 2) how can the transformation matrix $\mathbf{H}$ be ensured to be stable if a triangle is too misshapen? Here, we define that a misshapen triangle means that the minimum angle of the triangle is less than $\pi/6$, as shown in the magenta triangle in Fig. 5.

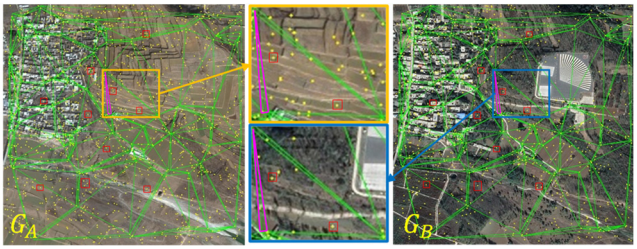



Fig. 5. Example of filtering out correct corresponding points. Some corresponding points are removed, as shown in the red rectangles. Magenta triangles represent the misshapen triangle.

In terms of mismatch points between two triangle meshes, we compare the similarity to determine the keypoints that matched. The similarity between the two triangles is higher, and the corresponding points are more reliable. The similarity between two triangles $\Delta ABC$ and $\Delta A'B'C'$ is $I$, the similarities between three corresponding angles are $I_a$, $I_b$, and $I_c$, the value of angle $A$ is $a$, and the value of angle $A'$ is $a'$. The formula for the similarity between angles $A$ and $A'$ is

$$I_a = \cos^3\left(\frac{\pi}{2}\left(1 - d\left(a'\right)\right)\right), \quad d\left(a'\right) = e^{-\frac{1}{2\sigma^2}\left(a'-a\right)^2} \tag{6}$$

where $\sigma = a/6$, the similarity $I_b$ and $I_c$ can be obtained in the same way, respectively, and the similarity of $\Delta ABC$ and $\Delta A'B'C'$ is

$$I = (I_a + I_b + I_c)/3. \tag{7}$$

The threshold $\delta$ is set at 0.75 to judge whether two triangles are similar. If the calculated similarity $I \geq \delta$, the two triangles are similar; otherwise, the two triangles are not identical.

In terms of the misshapen triangle, we set the minimum angle of the triangle greater than $\pi/6$ to void the misshapen triangle, which ensures that the matrix $\mathbf{H}$ is stable.

From the above analysis, the core of matching expansion based on the DT constraint is splitting the image plane into reliable transformations between two corresponding triangles and restricting the search space for candidate keypoint matches. Fig. 6 shows the process of matching expansion. For the DT $G_A$ and its corresponding graph $G_B$, nodes represent the feature keypoints extracted from images, and an edge is composed of a pair of adjacent keypoints. Two corresponding triangles $\Delta ABC$ and $\Delta A'B'C'$ are utilized to describe matching expansion, where $A$, $B$, and $C$, and $A'$, $B'$, and $C'$ are the corresponding point pairs. The affine transformation matrix $\mathbf{H}$ from triangle $\Delta ABC$ to triangle $\Delta A'B'C'$ is estimated using three nonlinear matching points. The second nearest neighbor feature keypoints inside triangles $\Delta ABC$ and $\Delta A'B'C'$ are denoted by $P$ and $Q$, respectively. The keypoint $p_i \in P$

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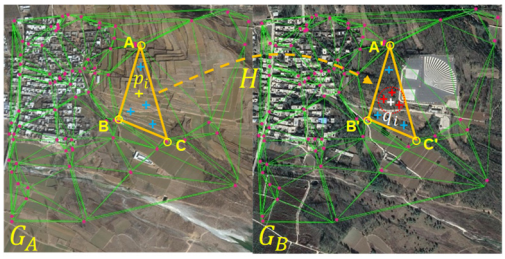



Fig. 6. DT constraint match expansion.

shown as a yellow cross line and its predicted location $q_e$ shown as a white cross line are computed by $\mathbf{H}*p_i$. Then, the keypoints inside the dashed red circle with center $q_e$ and radius $r$ are searched from $Q$, which are termed candidate keypoints of $p_i$. Similar to the one-way k-nearest neighbor keypoints sparse matching strategy, keypoint $p_i$ is compared with each candidate feature keypoints by computing the CNN descriptor Euclidean distance. The keypoint pair $(p_i, q_i)$ with the smallest distance, which is less than the specified threshold $t_d$, is labeled as an expanded corresponding point.

Algorithm 1 shows the matching expansion method based on the DT $G_A$ and its corresponding graph $G_B$ constraint strategy.

**Algorithm 1** Multimodal Image Matching With DT Strategy

**Input:** image pair $(I_A, I_B)$ with feature points P andQ; $n$ initial corresponding points $C_{init}$

**Output:** Final corresponding points $C_{fin}$

**Procedure**Delaunay triangulation

1: Extraction locations $L$ from feature points P based on initial corresponding points $C_{init}$

2: Construct the Delaunay triangulation $G_A$ using locations $L$ and its corresponding graph $G_B$ using candidate matches $C_{init}$

3: Outlier removal based on the dynamic adaptive distance constraint ($C_{dad} \leftarrow C_{init}$) according to eq.4 & eq.5

4: Corresponding point expansion based on the Delaunay triangulation constraint ($C_{exp} \leftarrow C_{dad}$)

**end procedure**

**Procedure** Corresponding point expansion

1: **for each** $t \in triangle(G_A)$

2: Calculate affine transformation matrix $H_{1,2}$ using three corresponding points based on two triangles with a similarity greater than 0.75 and a minimum angle greater than $\pi/6$ constraint

3: **for each** p = inside($t$)

4: Map a location $p_e = H_{1,2} * p$ in image $I_B$

5: Search candidate points $plist$ from the circular region around $p_e$ using threshold $t_d$

6: Determine the smallest Euclidean distance point $q$ of $p$ in $plist$

7: **if** $q \leq t_d$

8: Append an expanded corresponding point pair $(p, q)$

9: **end if**

10: **end for**

11: **end for**

12: Obtain $C_{fin}$ from refined $G_A$ and expanded corresponding point pairs

**end procedure**

## IV. Experimental Results and Analysis

All the experiments are performed on an NVIDIA GeoForce RTX 2080Ti GPU. To evaluate the superiority of M2DT-Net, we conduct experiments on seven different datasets to measure the performance of the proposed method, and the quantitative and qualitative results are compared with seven state-of-the-art methods.

### A. Experimental Datasets

*1) Datasets and Setups:* Although fine-tuning does not require as much data as training a deep CNN from scratch, a relatively large corpus of images is still considered a necessary prerequisite [48]. To solve the difficulty of acquiring abundant MRSIs in the same area with complex scenarios, we use computer vision image datasets with ground-truth labels for fine-tuning. The two disciplines, computer vision, and remote sensing, share a similar theoretical foundation, namely, the pinhole camera model and binocular vision principle in geometry [49]. Therefore, there are many overlaps in theories, methods, and techniques. Scholars also have many commonalities in image matching research. We suggest that the idea of matching in computer vision can be transferred to remote sensing image matching tasks.

In this study, model training includes a pretrained model and fine-tuning. We adopt the VGG16 architecture, pretrained on the *ImageNet* public training dataset, as the backbone network. The fine-tuning adopts *MegaDepth* [50] and *rotated HPatches* datasets. The *MegdDepth* dataset consists of 196 scenes from different viewpoints and was reconstructed from more than 10 million Internet photographs using COLMAP software. For each image, camera intrinsic and extrinsic parameters are provided. The *rotated HPatches* dataset is composed of the HPatches [51] and Google Earth [52] images; these images are rotated in-plane from 0° to 360° using a random rotation matrix. The network input is an image area cropped to a size of $512 \times 512 \times 3$. The model is fine-tuned by the SGD optimizer with a momentum of 0.9, a weight decay of 0.0001, and a base learning rate of 0.001. The network model is trained for 10k iterations.

*2) Test Dataset:* The test dataset consists of 21 pairs of MRSIs. These cases contain Optical–Optical (Pairs 1, 2, and 3), Optical–SAR (Pairs 4, 5, and 6), SAR–SAR (Pairs 7, 8, and 9), Optical–Lidar (Pairs 10, 11, and 12), Optical–Infrared (Pairs 13, 14, and 15), Infrared–Infrared (Pairs 16, 17, and 18), and Orthophoto–Oblique (Pairs 19, 20, and 21) images. These datasets contain MRSIs with various imaging mechanisms. The datasets are acquired from different sensors, resolutions, scales, times, and viewpoints, with different sizes and textures. In addition, the photographic area is also diverse in terrains, such as mountains, hills, and plains, and the test locations

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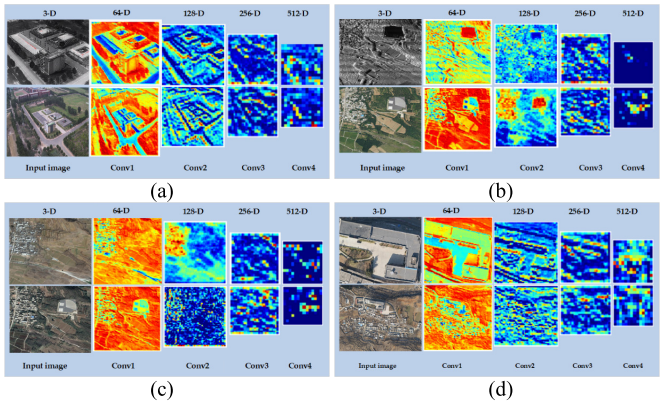



Fig. 7. Visualization results of learning features. (a) Infrared and optical images. (b) SAR and optical images. (c) Optical images at different times. (d) Optical images at different scales.

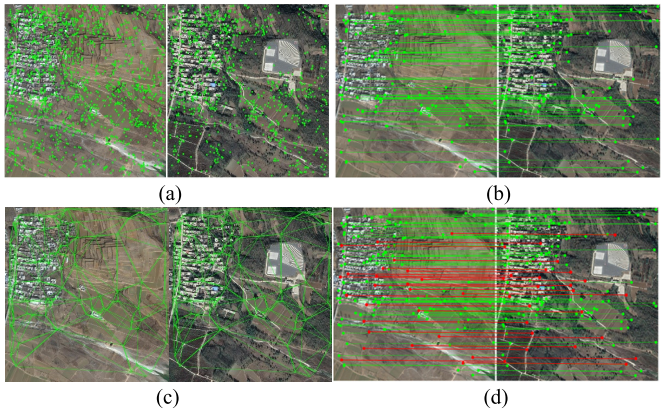



Fig. 8. Image matching using the DT constraint (Left: reference image and Right: sensed image). (a) Feature extraction. (b) Feature matching. (c) DT constraint. (d) Matching expansion.

contain suburbs and rural areas. A detailed description is shown in Table I.

*3) Evaluation Metrics:* To quantitatively evaluate the performance of the proposed method, the number of initial matches (NIM), the number of correct matches (NCM), the ratio of correct numbers (NCR), and the matching runtime are introduced as evaluation metrics to evaluate all methods. When the error rate of the matching point is less than 3 pixels, the point is considered a correct point. We report the average results of 3 runs to ensure the reliability of the experiments.

### *B. Visualization Results of Learning Features*

To better understand the characteristics of learning features in the MRSIs, we visualized different convolutional layers (Fig. 7). For the hierarchical CNN architecture, different CNN layers have learned different types of image learning features, and the bottom-up features form a hierarchical architecture. The bottom layer learns generic basic features, and the top layer extracts semantic features that are closer to the current task [53]. Section III-B describes the basic network architecture of the proposed learning feature extraction method. Fig. 7 shows the feature maps of the first layer to the fourth layer after convolution for various image matching cases. From these learning feature maps, the edge and texture features of the bottom feature maps are significant, and the top feature maps have gradually evolved into semantic features.

### *C. Analysis of DT Constraint*

Fig. 8 details the production of the proposed method. Fig. 8(a) shows the result of the feature extraction module, which generates abundant and evenly distributed feature keypoints. Fig. 8(b) shows the result of the feature matching module, which is aimed at removing outliers and obtaining correct corresponding points. Afterward, we construct the DT and its corresponding graph as similarly as possible using the locations from matched points according to the triangulation rules. Thus, the test image plane is divided into small patches to form the DT constraint presented in Fig. 8(c), which is the foundation of the matching expansion. Then, we further perform matching expansion based on the refined DT. Fig. 8(d) shows the expansion result, where the red line represents 26 underlying corresponding points after expansion.

### *D. Comparison With the State of the Arts*

To verify the effectiveness of the proposed algorithm, we conduct qualitative and quantitative contrast experiments on the seven MRSI datasets, as described in Table I. We compare M2DT-Net results with seven powerful methods on seven test datasets. The compared methods include RIFT, DELF, R2D2, D2-Net, SuperPoint, CMM-Net, and RoRD. Among them, RIFT is a handcrafted method with matches represented by yellow lines, while other methods are deep learning methods with matches represented by green lines in the qualitative results. For all the state-of-the-art methods that are compared, the matching results are obtained using the authors' released codes and models for a fair comparison.

*1) Qualitative Matching Result Comparisons:* Three image pairs from each MRSI dataset are selected for qualitative comparisons, as shown in Figs. 9–16. Among them, Fig. 9 suffers from very large differences in illumination and viewpoint. Fig. 10 has different imaging mechanisms and SAR images have obvious noise. Fig. 11 suffers from severe speckle noise and rotation changes. Fig. 12 has different imaging modes and the LiDAR image resolution is low. Fig. 13 suffers from large differences in viewpoint and imaging spectral band. Fig. 14 has very large differences in scale and viewpoint. Fig. 15 suffers from rotation, scale, time, and viewpoint differences. It follows that it is quite challenging to match these image pairs.

From the results, we can see that handcrafted RIFT based on a phase congruency map for feature detection and a maximum index map for feature description has translation invariance, but it is sensitive to rotation, scale, and speckle noise differences. For example, it is totally invalid in Figs. 9, 11, and 14. As analyzed in [54], the reason may be that speckle noise causes inaccurate edge structure information in the phase congruency map. For deep learning-based methods, DELF performs well in homologous optical or infrared images,

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TABLE I
DETAILED DESCRIPTION OF EXPERIMENT DATA

| Category | Pair | Dataset description | | |
|---|---|---|---|---|
| | | Reference image | Sensed image | Image characteristics |
| Optical -- Optical | 1 | Sensor: Optical<br>GSD: 1.0m<br>Size:1080×1080 | Sensor: Optical<br>GSD: 1.0m<br>Size:1080×1080 | Images cover a plain area with different seasons and viewpoints., having significant rotation differences. |
| | 2 | Sensor: Optical<br>GSD: 0.5m<br>Size:1386×925 | Sensor: Optical<br>GSD: 0.5m<br>Size:1073×785 | Images cover a village area with some buildings, having different illumination and viewpoint. |
| | 3 | Sensor: Optical<br>GSD: 0.5m<br>Size:1436×955 | Sensor: Optical<br>GSD: 1.0m<br>Size:1439×957 | Images cover a hills area with some buildings, having scale and viewpoint differences |
| Optical -- SAR | 4 | Sensor: Optical<br>GSD: 0.5m<br>Size:500×500 | Sensor: SAR<br>GSD: 0.5m<br>Size:500×500 | Images cover a hills area and have a temporal difference. Moreover, there is significant noise in the SAR image. |
| | 5 | Sensor: Optical<br>GSD: 0.5m<br>Size:500×492 | Sensor: SAR<br>GSD: 0.5m<br>Size:500×492 | Images cover an urban area and have a temporal difference. Moreover, there is significant noise in the SAR image. |
| | 6 | Sensor: Optical<br>GSD: 1.0m<br>Size: 1000×1000 | Sensor: SAR<br>GSD: 1.0m<br>Size: 1000×1000 | Images cover a suburban area and have a temporal difference. Moreover, there is significant noise in the SAR image. |
| SAR -- SAR | 7 | Sensor: SAR<br>GSD: 2.0m<br>Size: 600×500 | Sensor: SAR<br>GSD: 2.0m<br>Size: 600×500 | Images cover a plain area, having different viewpoints and sensors. There is significant noise in SAR images. |
| | 8 | Sensor: SAR<br>GSD: 0.5m<br>Size: 1500×1000 | Sensor: SAR<br>GSD: 1.0m<br>Size: 1500×1000 | Images cover a hills area with some buildings, and there is significant noise in the SAR image. |
| | 9 | Sensor: SAR<br>GSD: 1.0m<br>Size: 871×730 | Sensor: SAR<br>GSD: 0.5m<br>Size: 864×729 | Images cover a plain area, having different sensors. There is significant noise in SAR images. |
| Optical -- Lidar | 10 | Sensor: Optical<br>GSD: 0.5m<br>Size: 600×600 | Sensor: Lidar<br>GSD: 0.5m<br>Size: 600×600 | Images cover some buildings, and there is significant noise in the Lidar image. |
| | 11 | Sensor: Optical<br>GSD: 0.5m<br>Size: 600×600 | Sensor: Lidar<br>GSD: 0.5m<br>Size: 600×600 | Images are orthophoto images covering some buildings. There is significant noise in the Lidar image. |
| | 12 | Sensor: Optical<br>GSD: 0.5m<br>Size: 500×500 | Sensor: Lidar<br>GSD: 0.5m<br>Size: 500×500 | Images are orthophoto images covering some buildings. There is significant noise in the Lidar image. |
| Optical -- Infrared | 13 | Sensor: Optical<br>GSD: 0.5m<br>Size: 908×699 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 444×355 | Images cover some buildings. The infrared image is part of an optical image taken during the day. |
| | 14 | Sensor: Optical<br>GSD: 0.5m<br>Size: 681×512 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 640×512 | Images cover some buildings, and there is significant noise in the infrared image. It is taken during the night. |
| | 15 | Sensor: Optical<br>GSD: 0.5m<br>Size: 1403×950 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 640×512 | Images cover some buildings, and there is significant noise in the infrared image. It is taken during the night. |
| Infrared -- Infrared | 16 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 640×512 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 640×512 | Images cover some buildings and have viewpoint differences. There are taken during the day. |
| | 17 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 640×512 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 640×512 | Images cover a building and have scale differences. There are taken during the night. |
| | 18 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 640×512 | Sensor: Infrared<br>GSD: 0.5m<br>Size: 640×512 | Images cover a building and have a significant scale difference. There are taken during the night. |
| Orthophoto -- Oblique | 19 | Sensor: Optical<br>GSD: 0.5m<br>Size: 870×759 | Sensor: Optical<br>GSD: 0.5m<br>Size: 855×720 | Images cover a village with different scales, seasonal and light. The reference image is a digital orthophoto map. |
| | 20 | Sensor: Optical<br>GSD: 0.5m<br>Size: 870×759 | Sensor: Optical<br>GSD: 0.5m<br>Size: 957×784 | Images cover a village with different scales, seasonal and light. The reference image is a digital orthophoto map. |
| | 21 | Sensor: Optical<br>GSD: 0.5m<br>Size: 870×759 | Sensor: Optical<br>GSD: 0.5m<br>Size: 883×745 | Images cover a village with different scales, seasonal and light. The reference image is a digital orthophoto map. |

but it is not suitable for heterologous images, such as Optical–SAR, Optical–Infrared, and Optical–Lidar images. D2Net is the baseline of our method, it can be seen that it is only suitable for slight differences in scale, viewpoint, and imaging mechanism, as shown in Figs. 11, 13, and 14. R2D2 is robust to scale difference but easily influenced by viewpoint,

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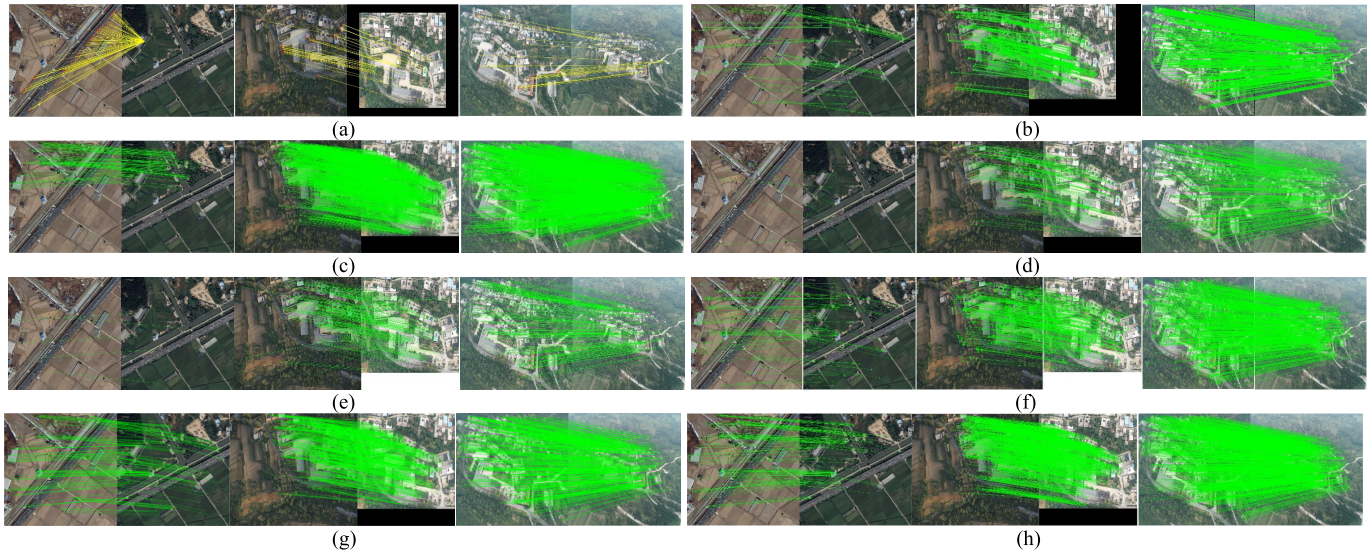



Fig. 9. Optical–Optical image matching results (from Left to Right, Pairs 1–3). (a) RIFT. (b) DELF. (c) D2Net. (d) R2D2. (e) SuperPoint. (f) CMM-Net. (g) RoRD. (h) M2DT-Net.

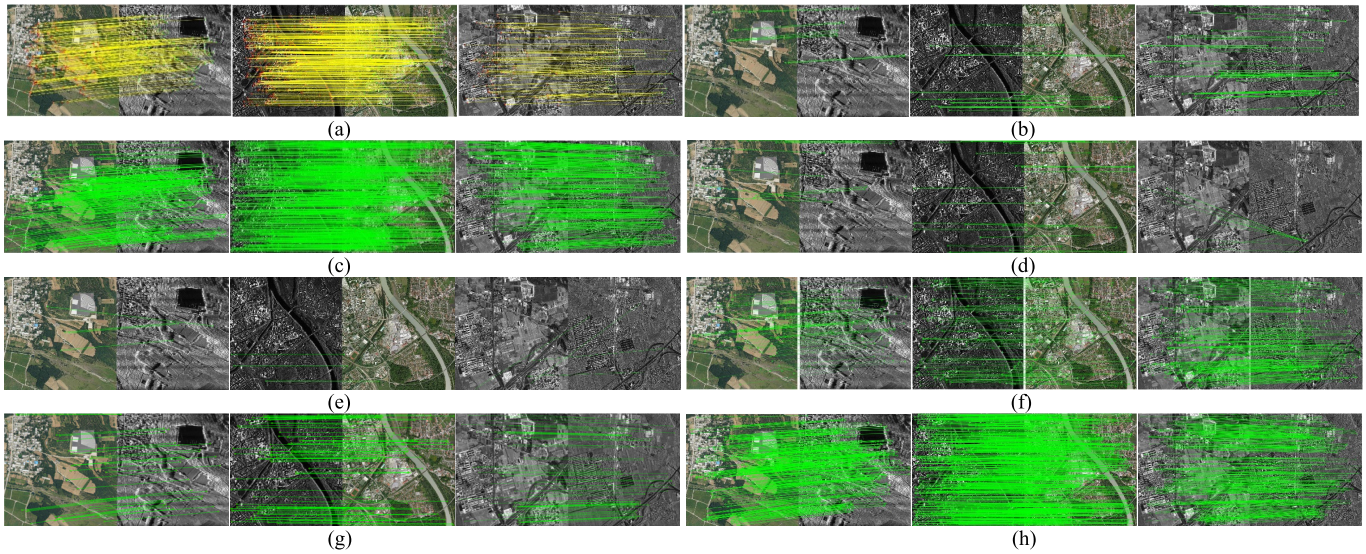



Fig. 10. Optical–SAR image matching results (from Left to Right, Pairs 4– 6). (a) RIFT. (b) DELF. (c) D2Net. (d) R2D2. (e) SuperPoint. (f) CMM-Net. (g) RoRD. (h) M2DT-Net.

rotation, and radiometric changes. SuperPoint is not suitable for MRSI matching with significant geometric and radiometric distortions because its training data are based on interest point detectors and descriptors. CMM-Net is stable for MRSI matching except for large viewpoint and scale changes, such as Figs. 13 and 14. RoRD is similar to the CMM-Net performance, which could suffer from slight differences in scale, viewpoint, and imaging mechanism.

In contrast, M2DT-Net achieves the best results on all image pairs. Visualization of qualitative matching results validates the superiority of the M2DT-Net. Moreover, our NCM is almost not affected by different image modalities. The key reason may be threefold: 1) we fine-tuned the baseline model to improve the MRSI feature extraction performance; 2) we combine adaptive distance constraint and DEGENSAC method to filter our outliers so as to retrain high-quality correspondence keypoints; and 3) we have carefully designed a match expansion strategy to find more underlying corresponding points so that they are suitable for MRSI matching.

*2) Quantitative Matching Result Comparisons:* Table II lists the average of three pairs of quantitative results on each dataset. For handcrafted RIFT method, it achieves more than 200 NCMs and 16% NCR on the Optical–SAR and Optical–Lidar datasets, which mainly suffer from radiometric distortions. Since the edge information in phase congruency is hard to handle large geometric distortions, it is sensitive to large-scale and viewpoint differences, validating in Optical–Optical, Infrared–Infrared, and Orthophoto–Oblique datasets.

For deep learning-based SuperPoint and DELF methods, they could not identify more correct matches compared with

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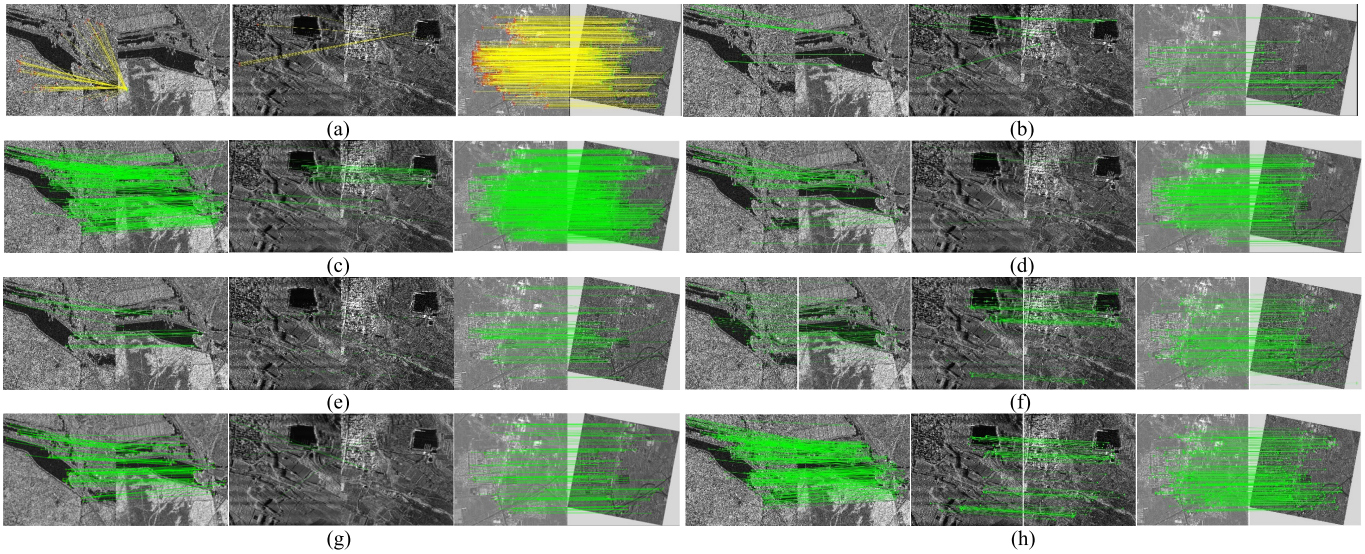



Fig. 11. SAR-SAR image matching results (from Left to Right, Pairs 7–9). (a) RIFT. (b) DELF. (c) D2Net. (d) R2D2. (e) SuperPoint. (f) CMM-Net. (g) RoRD. (h) M2DT-Net.

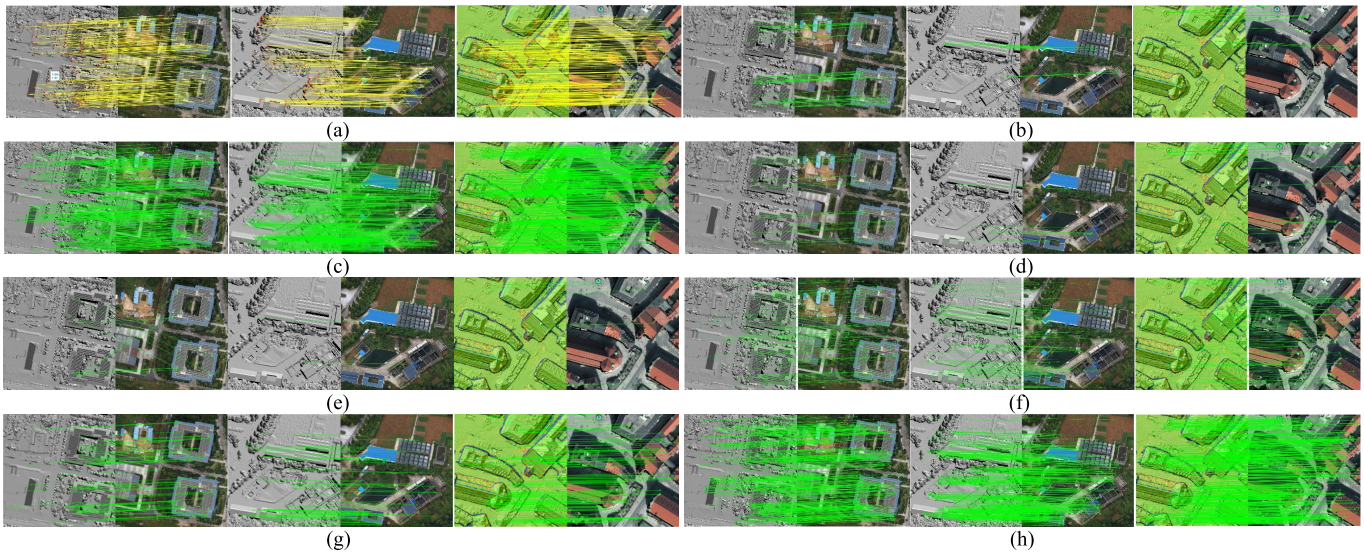



Fig. 12. Optical–Lidar image matching results (from Left to Right, Pairs 10–12). (a) RIFT. (b) DELF. (c) D2Net. (d) R2D2. (e) SuperPoint. (f) CMM-Net. (g) RoRD. (h) M2DT-Net.

other methods in most datasets, resulting in inferior matching performance on NCM and NCR. The lowest NCM and NCR of SuperPoint are 2 and 0%, respectively, and the highest NCR is still less than 10%. Namely, it almost fails on all the image pairs. DELF gets better results than SuperPoint, but its NCM and NCR are still low. It is excepted since they are not designed for multimodal image matching. Therefore, they are not suitable for MRSI matching tasks.

Since the difference in imaging mechanisms, salient feature keypoints are usually not covisible [55]. It can be seen that R2D2 and RoRD are less competitive due to fewer NCMs and NCRs on the Optical–SAR dataset. RIFT, D2Net, CMM-Net, and M2DT-Net have more abundant NCMs, all of which reach more than 223, indicating the effectiveness of these algorithms. Compared to its counterpart D2-Net, the NCR and time of our M2DT-Net are improved by more than 4% and 54.7%, respectively, and the NCM decreases slightly. Therefore, our model is much more efficient and lightweight.

Regarding viewpoint differences in Optical–Optical, Optical–Infrared, Infrared–Infrared, and Orthophoto–Oblique datasets, deep learning-based methods obtain more corresponding points. The main reason is that the training dataset has some different perspective changes, which increases the robustness of the network model to viewpoint differences. In addition, for large-scale differences, almost all methods are out of work, with the exception of M2DT-Net and R2D2, indicating that image multiscale features have a key role in improving the model matching performance with large-scale differences. These results further denote that M2DT-Net has good adaptability and stability to large-scale and viewpoint changes.

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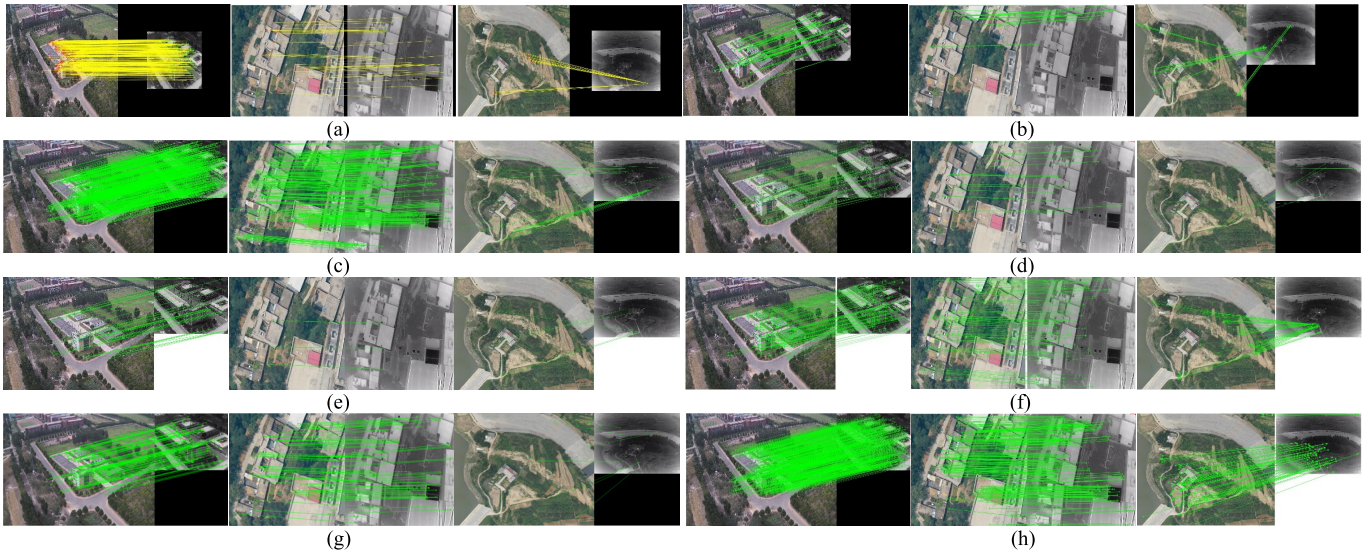



Fig. 13. Optical–Infrared image matching results (from Left to Right, Pairs 13–15). (a) RIFT. (b) DELF. (c) D2Net. (d) R2D2. (e) SuperPoint. (f) CMM-Net. (g) RoRD. (h) M2DT-Net.

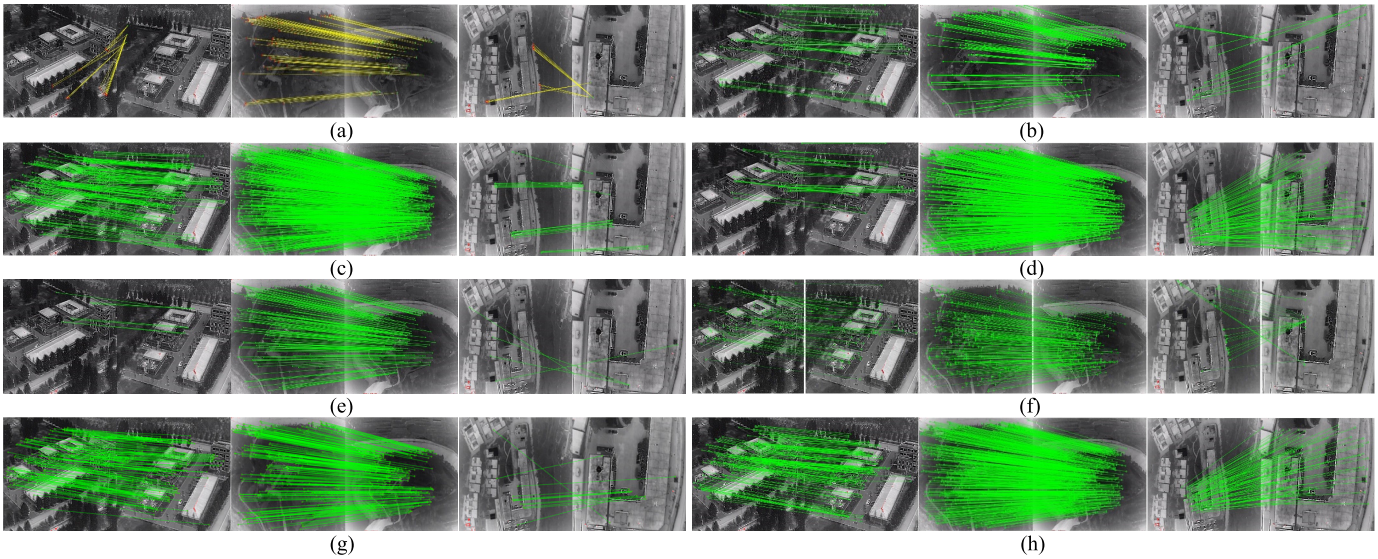



Fig. 14. Infrared–Infrared image matching results (from Left to Right, Pairs 16–18). (a) RIFT. (b) DELF. (c) D2Net. (d) R2D2. (e) SuperPoint. (f) CMM-Net. (g) RoRD. (h) M2DT-Net.

Overall, although the number of corresponding keypoints of deep learning methods is various, they can obtain superior results with nonlinear geometric and radiometric distortions. It indicates that deep learning-based methods are generalizable from nature images to remote sensing images.

The average NCR from lowest to highest of all methods on all the seven datasets is SuperPoint (3%), RIFT (7%), DELF (7%), CMM-Net (23%), RoRD (25%), R2D2 (30%), D2Net (40%), and M2DT-Net (44%). Among them, our M2DT-Net improves by 4%, 14% compared with D2Net and RoRD, and more than 20% compared with other methods. Compared to its counterpart D2Net, although the average NIM and NCM of the proposed M2DT-Net are inferior, the number decreased by 403 and 62, and the number of M2DT-Net is suitable for many multisource remote sensing applications. In addition, the average NCR and time are improved by over 4% and 14.76s, respectively, validating the usefulness of matching expansion.

The above experimental results prove that M2DT-Net can be applied to various types of MRSI matching tasks with rotation, scale, viewpoint, and radiation differences. Compared with a handcrafted method and various learning methods, our method achieves better performance and finds more correspondence points than other methods on the seven test datasets.

### *E. Ablation Study*

We conducted an ablation study by comparing the results on the MRSI datasets of M2DT-Net, where one of the algorithmic components is omitted or changed. This comparison allows us to evaluate the contribution of each component of the proposed

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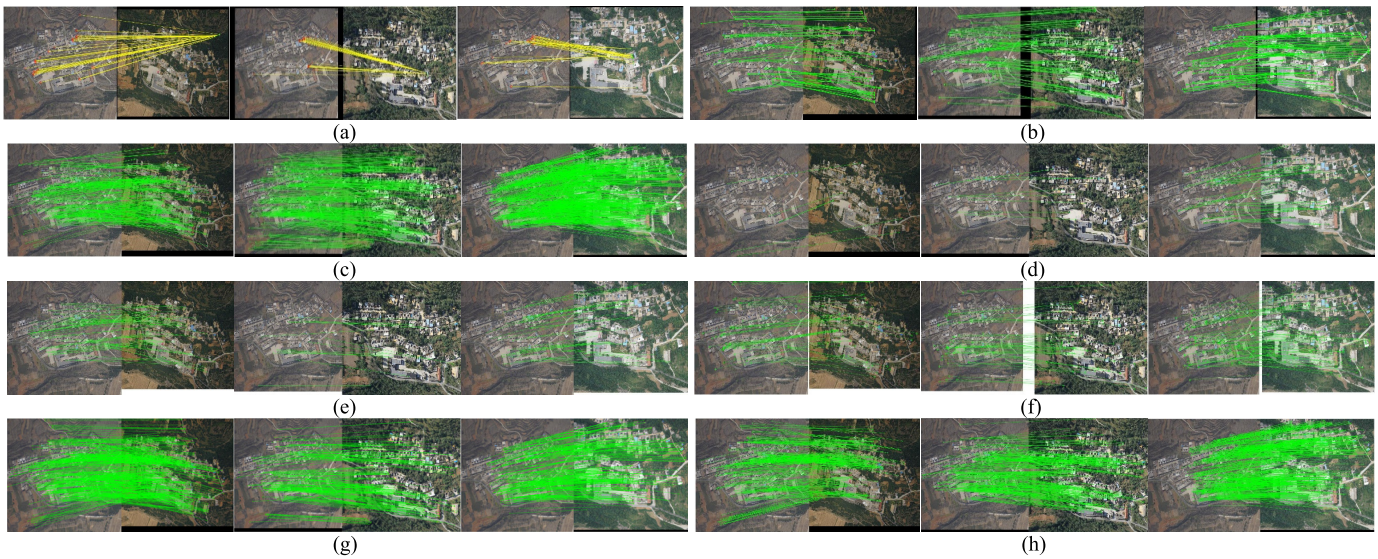



Fig. 15. Orthophoto–Oblique image matching results (from Left to Right, Pairs 19–21). (a) RIFT. (b) DELF. (c) D2Net. (d) R2D2. (e) SuperPoint. (f) CMM-Net. (g) RoRD. (h) M2DT-Net.

TABLE II
AVERAGE OF THE EXPERIMENTAL RESULTS

| Datasets | Metrics | RIFT | DELF | D2Net | R2D2 | SuperPoint | CMM-Net | RoRD | M2DT-Net |
|---|---|---|---|---|---|---|---|---|---|
| Optical - Optical | NIM | 1806 | 2015 | 2855 | 327 | 2475 | 1633 | 1771 | 2069 |
| | NCM | 16 | 191 | 1191 | 218 | 131 | 477 | 539 | 964 |
| | NCR (%) | 1% | 9% | 36% | 48% | 4% | 26% | 29% | 40% |
| | Time (s) | 5.62 | 5.98 | 58.9 | 5.81 | 1.49 | 4.34 | 6.94 | 9.35 |
| Optical - SAR | NIM | 1577 | 593 | 1185 | 209 | 1759 | 818 | 560 | 915 |
| | NCM | 254 | 27 | 487 | 13 | 4 | 223 | 75 | 438 |
| | NCR (%) | 16% | 6% | 50% | 6% | 0% | 34% | 17% | 54% |
| | Time (s) | 2.99 | 5.63 | 18.15 | 5.59 | 1.5 | 3.19 | 5.83 | 8.22 |
| SAR - SAR | NIM | 1425 | 352 | 1234 | 282 | 2039 | 1069 | 273 | 1071 |
| | NCM | 172 | 23 | 294 | 119 | 36 | 114 | 62 | 244 |
| | NCR (%) | 9% | 10% | 40% | 28% | 5% | 20% | 29% | 45% |
| | Time (s) | 4.77 | 2.24 | 34.71 | 5.75 | 1.5 | 6.74 | 2.83 | 9.43 |
| Optical - Lidar | NIM | 1348 | 559 | 712 | 116 | 2171 | 560 | 345 | 669 |
| | NCM | 218 | 20 | 266 | 18 | 2 | 160 | 84 | 366 |
| | NCR (%) | 16% | 3% | 52% | 15% | 0% | 31% | 24% | 56% |
| | Time (s) | 2.65 | 2.06 | 7.64 | 5.46 | 1.56 | 2.85 | 5.77 | 7.58 |
| Optical - Infrared | NIM | 1393 | 450 | 765 | 84 | 1407 | 773 | 289 | 468 |
| | NCM | 103 | 19 | 266 | 23 | 17 | 87 | 70 | 231 |
| | NCR (%) | 8% | 4% | 41% | 26% | 1% | 14% | 22% | 43% |
| | Time (s) | 3.13 | 2.23 | 8.6 | 5.38 | 1.47 | 3.12 | 5.81 | 8.66 |
| Infrared - Infrared | NIM | 1266 | 471 | 744 | 413 | 971 | 427 | 444 | 528 |
| | NCM | 25 | 50 | 309 | 318 | 58 | 123 | 132 | 250 |
| | NCR (%) | 2% | 11% | 41% | 77% | 10% | 28% | 29% | 47% |
| | Time (s) | 2.56 | 2.15 | 7.11 | 6.26 | 1.44 | 2.94 | 5.77 | 7.79 |
| Orthophoto - Oblique | NIM | 1762 | 1496 | 1881 | 127 | 1789 | 1067 | 1242 | 1250 |
| | NCM | 0 | 69 | 400 | 18 | 34 | 85 | 321 | 287 |
| | NCR (%) | 0% | 5% | 21% | 12% | 2% | 8% | 26% | 23% |
| | Time (s) | 3.81 | 2.86 | 28.98 | 5.58 | 1.5 | 3.67 | 6.8 | 9.72 |
| Average | NIM | 1511 | 848 | 1339 | 223 | 1802 | 907 | 703 | 996 |
| | NCM | 113 | 57 | 459 | 104 | 40 | 181 | 183 | 397 |
| | NCR (%) | 7% | 7% | 40% | 30% | 3% | 23% | 25% | 44% |
| | Time (s) | 3.65 | 3.31 | 23.44 | 5.69 | 1.49 | 3.84 | 5.68 | 8.68 |

algorithm. Table III summarizes the findings of the ablation investigation. Note that the values in Table III are the average results for all the different test datasets.

In terms of the network fine-tuning module, *Google Earth* and *rotated HPatches* images are used to perform fine-tuning the network model, and the NIM and NCR are

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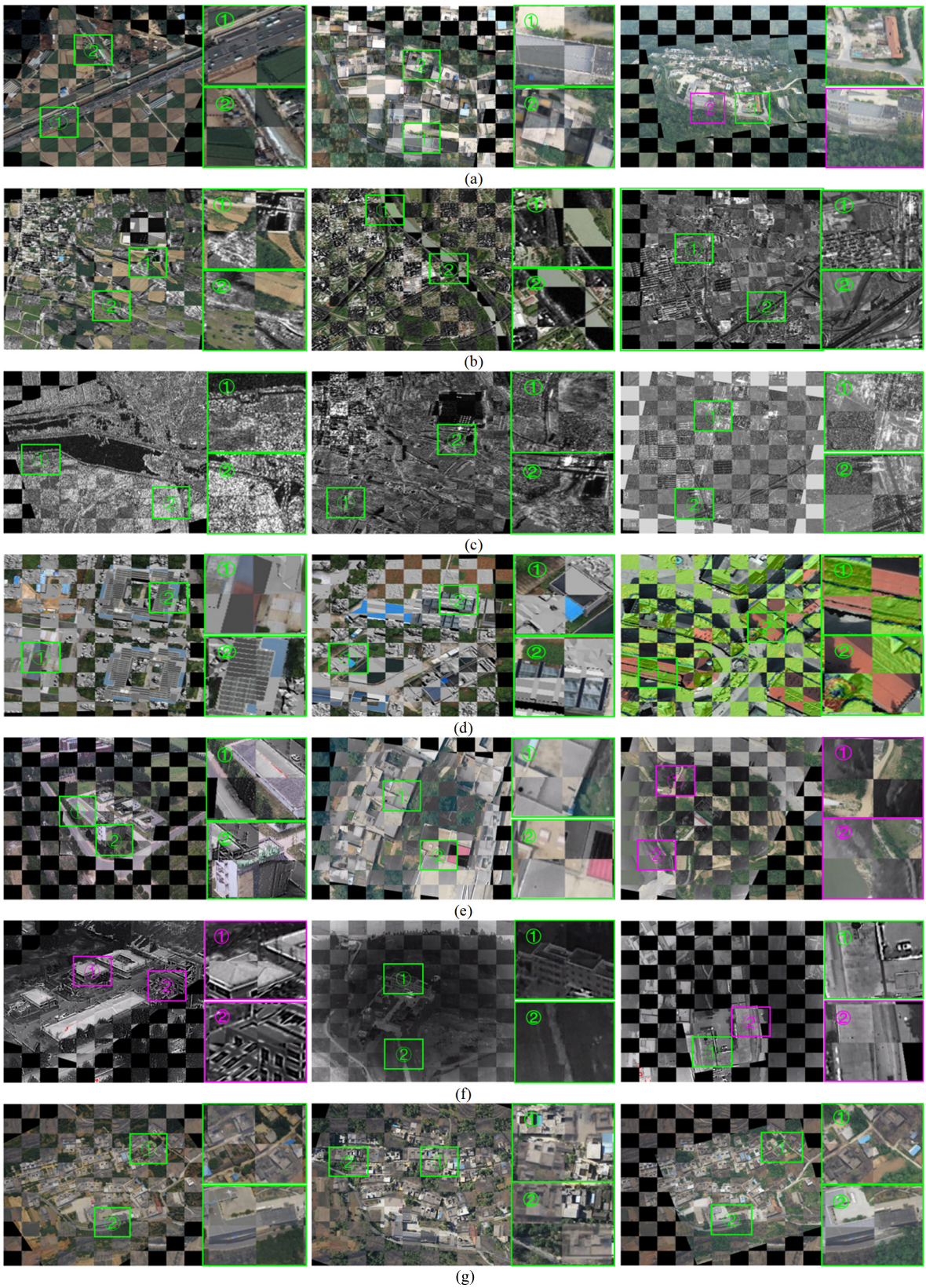



Fig. 16. MRSIs' registration results. (a) Optical–Optical image registration results. (b) Optical–SAR registration results. (c) SAR–SAR registration results. (d) Optical–Lidar registration results. (e) Optical–Infrared registration results. (f) Infrared–Infrared registration results. (g) Orthophoto–Oblique registration results.

significantly improved. The M2DT-Net method fine-tuned with a small amount of data can give a remarkable performance improvement on the MRSI dataset.

The multiscale feature extraction module can not only extract stable keypoints when there are obvious differences in scale, viewpoint, time, and imaging mechanism among the

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TABLE III
ABLATION STUDIES TO TEST THE ABILITY OF EACH MODULE OF M2DT-NET

| Method | Datasets | NCM | NCR | Time(s) |
|---|---|---|---|---|
| M2DT-Net w/o fine-tuning | Optical-Optical | 492 | 0.27 | 4.18 |
| | Optical-SAR | 218 | 0.33 | 3.05 |
| | SAR-SAR | 125 | 0.21 | 3.68 |
| | Optical-Lidar | 171 | 0.33 | 2.68 |
| | Optical-Infrared | 130 | 0.20 | 2.86 |
| | Infrared-Infrared | 137 | 0.36 | 2.59 |
| | Ortho-Oblique | 90 | 0.08 | 3.51 |
| M2DT-Net w/o multi-scale | Optical-Optical | 220 | 0.05 | 3.99 |
| | Optical-SAR | 119 | 0.14 | 3.18 |
| | SAR-SAR | 81 | 0.09 | 3.59 |
| | Optical-Lidar | 165 | 0.16 | 2.65 |
| | Optical-Infrared | 110 | 0.07 | 2.98 |
| | Infrared-Infrared | 101 | 0.13 | 2.69 |
| | Ortho-Oblique | 16 | 0.01 | 3.47 |
| M2DT-Net w/o DAD constraint | Optical-Optical | 196 | 0.29 | 4.52 |
| | Optical-SAR | 31 | 0.46 | 3.08 |
| | SAR-SAR | 32 | 0.27 | 4.11 |
| | Optical-Lidar | 18 | 0.50 | 2.95 |
| | Optical-Infrared | 14 | 0.41 | 3.21 |
| | Infrared-Infrared | 86 | 0.43 | 2.97 |
| | Ortho-Oblique | 9 | 0.28 | 3.67 |
| M2DT-Net w/o DT constraint | Optical-Optical | 798 | 0.33 | 4.18 |
| | Optical-SAR | 359 | 0.44 | 3.05 |
| | SAR-SAR | 193 | 0.35 | 3.68 |
| | Optical-Lidar | 284 | 0.43 | 2.68 |
| | Optical-Infrared | 180 | 0.34 | 2.86 |
| | Infrared-Infrared | 188 | 0.33 | 2.59 |
| | Ortho-Oblique | 240 | 0.19 | 3.51 |
| M2DT-Net full | Optical-Optical | 964 | 0.40 | 9.35 |
| | Optical-SAR | 438 | 0.54 | 8.22 |
| | SAR-SAR | 244 | 0.45 | 9.43 |
| | Optical-Lidar | 366 | 0.56 | 7.58 |
| | Optical-Infrared | 231 | 0.43 | 8.66 |
| | Infrared-Infrared | 250 | 0.47 | 7.79 |
| | Ortho-Oblique | 287 | 0.23 | 9.72 |

MRSIs but also integrate multilevel local feature information, rendering the feature keypoint description more robust and more distinguishable when faced with apparent geometric and radiation distortions.

In terms of the DAD constraint module, compared with the FLANN match where the threshold of good match points is set to 0.9 and only a few correspondence points are obtained, our method obtains abundant corresponding points on the different MRSI datasets. The reason for this result is that the DAD module can set appropriate thresholds for different cases rather than giving an absolute ratio. This finding certifies the superiority of the DAD constraint strategy.

The NCM and NCR metrics are significantly improved by fine-tuning the pretrained model, extracting the multiscale features, and adopting the DAD constraint strategy. However, there are still some underlying correspondence points, which are suppressed by nonsignificant features. It can be observed that the proposed M2DT-Net fully improves NCM and NCR by approximately 20.2% and 21.3%, respectively, compared with M2DT-Net without the DT constraint module, indicating the effectiveness of our DT constraint strategy.

More interestingly, when all algorithmic components are combined, our method achieves the highest NCM and NCR values. Based on the aforementioned analysis, we conclude that each component is critical for M2DT-Net. These observations lead the proposed method toward more practical serviceability.

### *F. Results on Image Registration*

To further evaluate the effects of our method, we applied the proposed M2DT-Net to register MRSI tasks. Fig. 16 shows the qualitative registration results in the chessboard mosaic image, where green rectangles show consistent alignment and magenta rectangles represent inconsistent alignment. We observe that the edges of almost all ground objects are continuous and that the overall area overlaps with our results. The reason is that the proposed method can obtain evenly distributed and sufficient matches, which are more conducive to the calculation of the image transformation parameters. However, for the very large-scale and viewpoint differences in Pairs 3, 15, 16, and 18, a small number of regions appear discontinuous under the influence of nonlinear radiometric differences and regional geometric feature variations. In fact, these image pairs have great content changes.

## V. CONCLUSION AND FUTURE WORK

This article proposes an MRSI matching method named M2DT-Net that combines learning features and the DT constraint. The main contribution is that M2DT-Net is suitable for various MRSI high-precision matching under complex real-world scenarios, achieving a balance of robustness and time consumption. The proposed method consists of three modules: a feature extraction module, a feature matching module, and a matching expansion module. The feature extraction module extracts abundant local features; the feature matching module retains high-quality matching points; the matching expansion module realizes as many inliers as possible. According to the qualitative and quantitative results, our method obtains good matches in all test scenarios and has advantages in NCMs and NCRs metrics. Compared with the other seven state-of-the-art methods, our method has greater distinctive and unambiguous adaptability and superiority in the number of corresponding points, matched points distribution, and matching efficiency.

However, considering that the deep learning features are data-driven feature representations, sufficient training data are required if we want to obtain a CNN model with strong performance. Therefore, the idea of transfer learning can be further exploited to build more robust models and increase the annotation of the target data to achieve better MRSI matching. In future work, we will focus on improving the matching precision and efficiency for few-shot matching, large oblique image subpixel accuracy matching, and more prior information constraints.

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